



Exploring auditory feedback mechanisms in speech recognition

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Abstract

For many years, automatic speech recognition (ASR) has been built on compressed filter-bank features understood to be a rough model of the cochlea. However, recent understanding, evidenced by oto-acoustic emissions, is that the cochlea is composed of driven oscillators. The Hopf mechanism arising from an oscillator model explains the well known cube-root compression. A bifurcation arises from an inner feedback loop from the outer to inner hair cells. Further, larger feedback loops exist along the efferent path of the auditory nerve, particularly from the olivary complex. The means of combination of these signals is less well understood. In the present study, to the extent to which current compute power allows, we investigate how to incorporate the Hopf mechanism and the olivocochlear feedback mechanisms into ASR. Results show that adding this larger feedback loop appears beneficial for ASR. The results currently have modest implications for ASR, however, perhaps more importantly, such technology could be used to make inference about the biological mechanism.

Index Terms: speech recognition, Hopf oscillators, cochlear model, efferent pathway

1. Introduction

For several years, cochlear models have been a topic of research in both neuroscience and automatic speech recognition (ASR). Although these fields differ in computational objectives, they share a common underlying process: transforming human speech into comprehensible signals, whether they be neural signals, phonemes, or words. Physiological models first served as a foundation for ASR technology. However, driven by computational constraints and performance, the ASR field has increasingly focused on filter-based speech features such as Mel-Frequency Cepstral Coefficients (MFCCs), or more recently raw speech with end-to-end (E2E) processing [1].

In the physiological field, cochlear models are generally designed to better understand the underlying human mechanism. Based on these studies, the mathematical modelling that currently best approaches the intricacies of the cochlea is the Hopf oscillator model [2, 3]. This model offers a filtering aspect as well as a damping regime and an active amplification regime. Moreover, in the auditory pathway from the cochlea to the brain, several efferent paths have been reported. The best-known is the olivocochlear feedback mechanism from the olivary complex (OC) to the cochlea [4, 5]. This mechanism protects against damage from loud noise, improves speech perception in noisy environments, and modulates the auditory nerve's sensitivity.

Hopf oscillators have already been shown to be capable of performing sound recognition [6]. In this paper we investigate the incorporation of a Hopf oscillator frontend model and an ef-

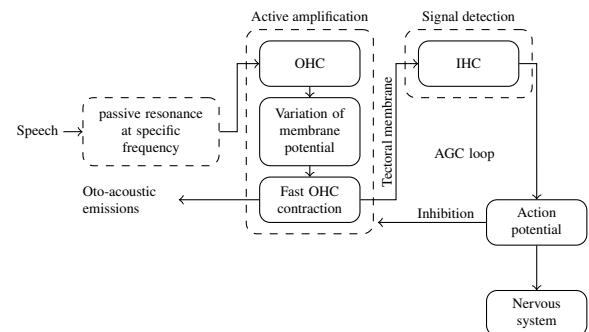


Figure 1: Schematic of the working of the cochlea

ferent path in an ASR context. We show that the Hopf oscillator is capable of generating comprehensive features and moreover, the efferent path enhances the global ASR performance.

2. Background

2.1. Speech features in ASR

In traditional ASR systems, MFCCs are commonly employed as feature extraction. MFCCs serve as a simplified representation of the cochlea's function. Sample windows of 25ms are extracted from a raw speech signal (the sampling rate is typically about 16kHz) at a 100Hz frame rate which constitutes frames. Those frames are further processed, which involves a logarithmic compression of the amplitude to reduce dynamic range and mel-frequency warping to align with human hearing sensitivity.

By contrast, modern ASR systems often employ raw speech as input [1], leveraging convolutional neural networks to extract meaningful features. This shift has resulted in improved performance, yet at the cost of reduced interpretability.

2.2. Physiological working of the cochlea

The working of the cochlea was first understood to function as a passive filterbank [7]. However, some studies indicated that a passive filter system would not be capable of handling the amplitude range of the human ear [8], which implies the need for an active amplification mechanism to handle low-amplitude signals. A better understanding of the cochlea has revealed that it functions as an array of active amplification oscillators. Figure 1 illustrates the inner workings of the cochlea, showing how the outer hair cells (OHCs) detect small movements of the basilar membrane and actively amplify them. Conversely, when the inner hair cells (IHCs) detect these movements, they generate a neural signal that is further transmitted through the nervous sys-

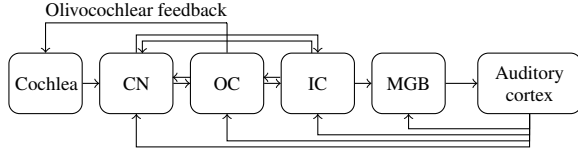


Figure 2: Auditory path from cochlea to midbrain with the efferent paths from the auditory cortex and midbrain.

tem to the brain. An inhibitory feedback loop to the OHCs regulates their amplification activity. The electromotility capacity of OHCs is also responsible for generating oto-acoustic emissions (OAEs). In a silent environment, the OHCs oscillate, producing vibrations that can propagate backwards through the ear and generate sound [9, 10, 11].

2.3. Cochlear models

Cochlear models have been developed to capture various aspects of the cochlea's functioning, each focusing on distinct aspects. These models aim to better understand the complex interactions within the auditory system, including the relationships between neurons, synapses, and the fluid dynamics of the basilar membrane. One type of model concentrates on the shape of the impulse response, such as gamma-tone filters, which exhibit useful capacities for applications like ASR and cochlear implants [12, 13, 14, 15]. Another type of model focuses on biochemical transfers in synapses, with neurotransmitter-based cochlear models ultimately leading to the generation of spikes [16, 17]. A third type models the basilar membrane and underlying fluid dynamics. These models focus on the transmission of sound vibrations through the perilymph and the tonotopic distribution of frequencies in the cochlea [18, 19, 20]. A fourth type are based on the active amplification mechanism inherent to the OHC-IHC interactions. Notable examples are CARFAC [21, 22] and the Hopf oscillator mechanism [2, 3].

2.4. The efferent path

Beyond the cochlea, several efferent paths have been reported to play important roles in modulating auditory perception [23] (see Figure 2). One well-studied efferent path is the olivocochlear feedback, which connects from the OC to the cochlea [4, 5]. This pathway is present in various mammals and has been favoured by evolutionary selection [24, 25]. However, it's not the only efferent path; other descending projections from the auditory cortex to the cochlear nerve (CN), OC, inferior colliculus (IC) and medial geniculated body (MGB) play an important role in synapse adaptation. Furthermore, IC, SOC, and CN are interconnected within a bidirectional network that facilitates top-down modulation in the auditory pathway.

3. Hopf oscillator formulation

3.1. Mathematical formulation

The cochlea acts an active amplification oscillator. A mathematical formulation can be derived from a harmonic oscillator describing the stereocilia movements by adding an active force component f_a [26].

$$m\ddot{x} + h\dot{x} + kx = F + f_a \quad (1)$$

Where x is the displacement and m , h and k are constants. However, the mathematical development leading to a Hopf os-

cillator is complex [3]. A simpler approach is to start from the normal form of the Hopf oscillator in the polar domain [27]:

$$\begin{cases} \dot{r} = r(\mu - r^2) \\ \dot{\theta} = 2\pi f \end{cases} \quad (2)$$

This equation consists of two differential equations: the phase (θ) is directly dictated by the oscillator frequency (f), while the radius (r) equation yields a more complex behaviour that includes a cube root compression and a pitchfork bifurcation tuning through the parameter μ . This bifurcation parameter (μ) defines the number of solutions of the differential equation and consequently the bifurcation point.

- When $\mu < 0$, the equation has one stable solution: $r = 0$.
- When $\mu > 0$, the system has two stable solutions: $r = \pm\sqrt{\mu}$ and one unstable solution $\mu = 0$.
- $\mu = 0$ corresponds to the bifurcation point.

The Hopf oscillator model (equation 2) has been adapted for cochlear applications by incorporating additional features [28, 29, 3]. Firstly, an external signal parameter (x) is introduced to drive the cochlea by incoming speech signals. Secondly, a tuning equation for the bifurcation parameter is added to the model. This equation allows the system to transition from a damping regime (characteristic of loud sound levels) to an active amplification regime (more suitable for quiet situations) [2]. Thirdly, a frequency-scaling factor is applied to the differential equations. This parameter is proportional to the bandwidth of each oscillator [30, 31, 32].

In our model, the Hopf oscillator equations are defined through differential equations and bifurcation parameter μ is tuned through a feedforward mechanism dependent on the input signal x . These equations are summarized as:

$$\begin{cases} \dot{r} = (\mu r + \beta r^3 + x \cos \theta) \pi f_{bw} \\ \dot{\theta} = 2\pi f_c - \frac{x}{r} \pi f_{bw} \sin \theta \\ \mu_{th} = \mu_{max} \left(\frac{x^2}{\delta^2} - 1 \right) \end{cases} \quad (3)$$

Where the different parameters are defined as:

- β The first Lyapunov coefficient. It needs to be negative in order to have a stable solution over the whole domain.
- μ_{max} The maximum value fixed for the bifurcation parameter.
- δ The bifurcation threshold, defining the limit between the damping and active amplification mode.
- f_c, f_{bw} Respectively the central frequencies and bandwidth of the different oscillators defined based on the mel-scale.

3.2. Characterization

The Hopf module is designed so that a cube root equation governs the evolution of the oscillator radius as a function of the input signal amplitude, with the bifurcation parameter fixed at $\mu = 0$:

$$r = \sqrt[3]{-\frac{A}{2\beta}} \quad (4)$$

Since the bifurcation parameter is able to adapt to the incoming signal, the equilibrium radius curve as a function of the input signal amplitude will slightly diverge from the initial cube root expression illustrated in Figure 3. The threshold between the two regimes happens when:

$$A = \sqrt{2}\delta \quad (5)$$

For low amplitudes, the bifurcation parameter is above zero, which actively amplifies the radius and for high amplitudes the

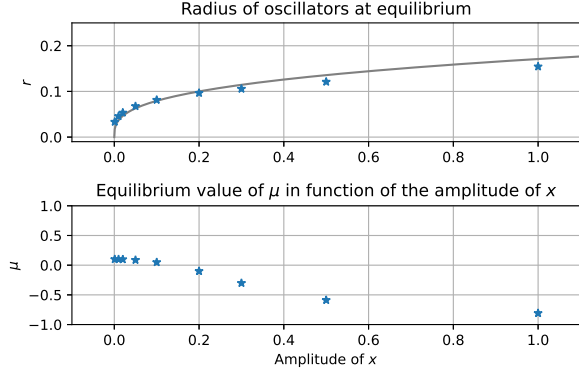


Figure 3: *Diverging tendencies from the cube root relationship between the input signal amplitude (A) and the oscillator radius (r).*

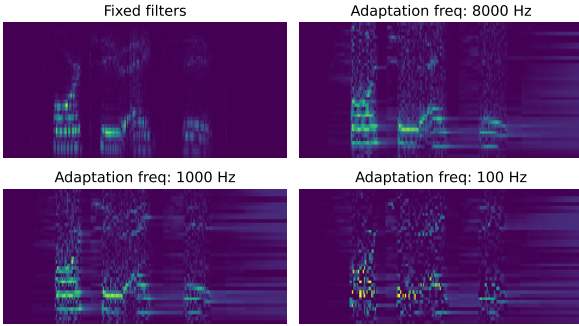


Figure 4: *Illustration of the fixed filter output and the oscillator output for different Hopf adaptation frequencies.*

bifurcation parameter gets lower which damps down the oscillator radius.

3.3. Computational trade-off of Hopf frequency rate

The system parameters are updated using the Euler step for the differential equations, while the bifurcation parameter is updated through a proportion-wise combination with its previous value. To prevent divergence, a tanh function is applied to the computed bifurcation parameter:

$$\begin{cases} r[t + \Delta t, t + 2\Delta t] = r[t, t + \Delta t] + \dot{r} \cdot \Delta t \\ \theta[t + \Delta t, t + 2\Delta t] = (\theta[t, t + \Delta t] + \dot{\theta} \cdot \Delta t) \% 2\pi \\ \mu[t + \Delta t, t + 2\Delta t] = p\mu + (1 - p) \tanh(\mu_{th}) \end{cases} \quad (6)$$

An Euler step update implies using a recurrent implementation. Note that the Euler updates proceed at sample rate rather than the more usual frame rate, implying a computational overhead. Recurrence at sample rate leads to training times measured in weeks on modest current graphics units. Hence from the outset we seek to mitigate recurrence frequency. The Hopf adaptation rate can be lowered to different frequencies, which results in larger Δt steps in equation 6 to compute the Hopf parameters. Reducing the adaptation frequency improves computational efficiency but increases signal distortion. A reasonable lower bound for preserving speech features is the frame rate. Figure 4 compares the output of a fixed filter with that of a

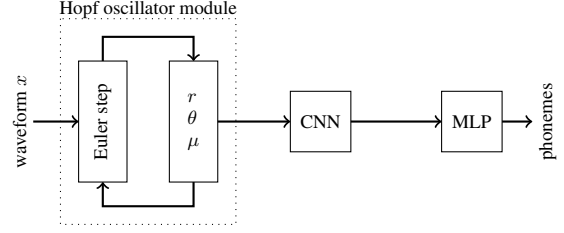


Figure 5: *Integration of Hopf module in a simple ASR baseline.*

Hopf oscillator under different adaptation frequencies. We observe a linear relationship between the Hopf adaptation rate and the computation time.

Figure 4 compares the fixed filter output with Hopf oscillator output with different Hopf adaptation frequencies. We note that there exists a linear relationship between the Hopf adaptation rate and the computation time.

4. Integration into ASR

The primary objective of this paper is to investigate the feasibility of integrating a Hopf oscillator model with an ASR system, as well as to evaluate the effectiveness of combining the efferent pathway feedback capacities with the Hopf oscillator model. To accomplish this, we utilize WER as a performance metric, which serves as a robust benchmark for comparing results across various configurations.

4.1. Learning paradigms

In ASR and other deep learning tasks, parameter optimization is typically achieved through gradient descent, a method that iteratively adjusts model weights to minimize prediction errors. This process shares similarities with neural adaptation, where synaptic strengths and types are shaped by neurotransmitter flow. In this context, the gradient descent mechanism can be seen as analogous to how these synaptic adjustments occur in biological systems. Unlike traditional gradient-based methods like gradient descent, the Hopf module uses differential equations to dynamically adjust parameters in real-time according to the incoming signal, even during testing.

4.2. Integration of the Hopf module

A first experiment consist of drawing a baseline using the Hopf oscillator module in an ASR system composed out of a CNN and an MLP as illustrated in Figure 5. This study leads to two main hypotheses. Firstly, that Hopf oscillators, more than only being the best model to capture some cochlear intricacies are able to generate features that an ASR system can use with a reasonable performance compared to fixed rectangular filters from SincNet [33]. Second, that the trade-off discussed in section 3.3 also relates to the time-performance trade-off in the ASR experiment. A lower adaptation rate would then imply a lower performance but a faster computation.

4.3. Integration of efferent path

The auditory path contains several efferent pathways from which the olivocochlear feedback is the best known (see section 2.4). This path connects the OC to the OHC and IHC. We propose to model a similar feedback loop by a transposed CNN. Therefore, we connect this feedback to the output of the Hopf

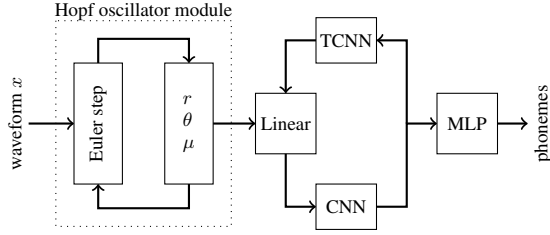


Figure 6: *Integration of efferent path in the ASR baseline combined with the Hopf oscillator frontend.*

oscillator through a linear module as illustrated on Figure 6. A second feedback mechanism originates from the auditory cortex and extends to various nuclei along the auditory pathway. Its primary function is to modulate synapses, allowing for the adjustment of incoming neural signals. This form of feedback is already modelled through gradient backpropagation, which directly influences the weights across different modules.

We hypothesize that incorporating a large feedback loop into the ASR structure, analogous to the olivocochlear feedback mechanism, could further improve overall ASR performance.

5. Results

We conducted the different experiments on the TIMIT dataset [34], utilizing the pytorch-kaldi framework [35]. We use 40 oscillators in the Hopf module, distributed over the mel frequency range [36, 37] and we perform the experiments on several adaptation frequencies between the sampling and frame rate. For the experiments involving a larger feedback loop, the feedback adaptation frequency is set to 50 Hz. The ASR performance achieved for the different experiments as well as the fixed filter baseline are illustrated on Figure 7.

5.1. Hopf oscillator baseline

The baseline results support our hypothesis that Hopf oscillators produce features that an ASR system can use for phoneme translation and that shows a comparable performance to fixed filters. Moreover, our findings indicate a clear trade-off between performance and efficiency, where lower adaptation frequencies result in reduced computational time, but at the cost of increased WER. This trade-off is confirmed by experiments with different frequency settings, which reveal that:

- A 100 Hz adaptation rate offers similar time efficiency to fixed filters, but the performance is only around 43% WER.
- With an adaptation frequency between 4 and 16 kHz, the performance of ASR models are similar to the performance obtained with fixed filters. The time per epoch however is multiplied by a factor between 10 and 50.

5.2. Hopf oscillator with feedback mechanism

The addition of a large feedback loop is, on the one hand, computationally more demanding, but on the other hand, it brings a performance improvement over all the tested configurations which demonstrates the hypothesis that a feedback mechanism would improve the global system performance.

- For Hopf adaptation frequencies below 500Hz, the baseline implementation provides a better efficiency-performance trade-off.
- For Hopf adaptation frequencies above 500Hz, the perfor-

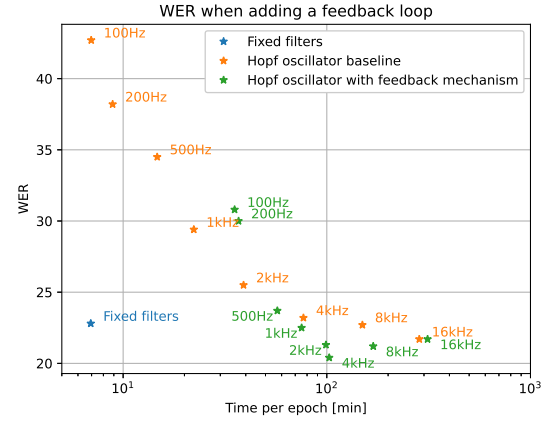


Figure 7: *WER results of the Hopf oscillator for different Hopf adaptation frequencies for both experiments with and without adding a large feedback loop.*

mance beats this trade-off and above 2kHz, it achieves even better results than the baseline at 16kHz.

This result indicates that the Hopf mechanism on its own does provide an output that requires an additional feedback mechanism, in the same way as the cochlea is better tuned with the efferent path. Hopf oscillators have the tendency of oscillating with a given inertia depending on the bifurcation parameter. This inertia without a proper feedback mechanism could be misinterpreted by the ASR system, working on a frame rate basis without any recurrent knowledge of previous frames.

6. Conclusion

In this paper, we investigate the working of the cochlea as a Hopf oscillator and evaluate its capacity in an ASR context. The inner workings of the cochlea are mathematically best represented by Hopf oscillators, which exhibit cube root compression and adaptation around the Hopf bifurcation. This bifurcation causes the mechanism to switch from an active amplification function to a damping mechanism based on the amplitude of the input signal. When incorporated into an ASR system, the module introduces two adaptation mechanisms: one driven by differential equations to dynamically adjust the parameters of the Hopf module, and another governed by an autograd system to optimize the weights of the ASR's various components. Furthermore, we incorporate larger auditory pathway feedback mechanisms into the ASR system, which mirrors the feedback process of the auditory brain.

The results from a Hopf integration experiment demonstrate that Hopf oscillators can produce speech features that can be processed by an ASR system. A trade-off curve shows that a high adaptation rate above 4kHz yields equivalent performance to fixed rectangular filters. The efferent path experiments reveal that incorporating this feedback mechanism significantly improves experimental performance, emphasizing the necessity to adapt Hopf oscillator outputs.

This outcome suggests that physiologically inspired adaptations can enhance the global functionality of physiologically plausible models. Developing this framework may lead to further insights about the auditory mechanism.

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8. References

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